

EDUCATION

- **Indian Institute of Information Technology, Lucknow** Lucknow
Bachelor of Information Technology; CGPA: 8.32 Apr 2020 - Apr 2024
- **Relevant Courses:** Data Structures & Algorithms, Operating Systems, System Design, DBMS, Object-Oriented Programming, Distributed Systems.

PROFESSIONAL SUMMARY

- Backend engineer with **3+ years** building scalable distributed systems, microservices, and APIs in production.
- Delivered measurable reliability and performance gains, including a **208× query speedup** on a 524M-row table and zero-downtime database migrations across **4 production regions**.
- Hands-on with **Java/Spring Boot, Ruby on Rails, SQL, Redis, ClickHouse, AWS (RDS), Jenkins, and CI/CD**, with strong ownership of deployments and operational excellence.

EXPERIENCE

- **BrowserStack** Mumbai (Mallard West)
SDE - Backend - Ruby, Rails, Express, JavaScript, SpringBoot, Java, ClickHouse, Helm, Jenkins Apr 2025 - Present
 - Achieved a **208× query speedup** (9,652ms → 46ms) on a **524M-row** sessions table via a composite covering index and zero-downtime **gh-ost** migration across **4 regions** (US, EU, IN, AUS).
 - Averted a projected production outage by executing zero-downtime **INT→BIGINT** migrations on **3 billion-row** tables (1TB each) growing **2.89M rows/day** across all regional RDS clusters.
 - Remediated critical **SQL injection** vulnerabilities across **50+ API endpoints** spanning 4 teams, securing a production system handling **4,000 req/s**.
 - Drove org-wide **Semgrep SAST** adoption for automated SQLi prevention at CI level, authoring rules covering **56 P0/P1 API actions** (**98%** of API traffic).
 - Resolved **2 P0** vulnerabilities (OS command injection/RCE and Axios SSRF bypass) on production endpoints before exploitation.
 - Expanded production health monitoring from **18 to 101 tracked endpoints** across **4 regions**, with per-endpoint p90 latency alerting and a **BigQuery** trend-analysis pipeline.
 - Cut primary-database load by migrating heavy aggregate queries from RDS to **ClickHouse**, re-architecting the Query Killer from cron-to worker-based execution, and reducing RDS master CPU from **96%** to stable levels.
 - Designed sharding for **35 Debezium MySQL CDC connectors** and executed Kafka history-topic splits across 3 regions, improving pipeline reliability and scalability.
- **Zus (formerly Ochain)** Cupertino, CA (Remote)
Software Engineer - Blockchain Storage & Mobile - Golang, Kotlin, GitHub Actions, Gin, Grafana Feb 2022 - Mar 2025
 - Architected and deployed two production Android apps: **Bolt** (crypto wallet) and **Vult** (decentralized storage), driving **20%** increase in platform adoption and processing **\$2M+** in transactions.
 - Spearheaded comprehensive **UAT automation suite** for enterprise blockchain storage, reducing critical production bugs by **90%** and cutting release cycles from **2 weeks to 3 days**.
 - Optimized **S3-compatible distributed storage** infrastructure handling **500M+ files** and **40TB+** data across **50+ nodes**, achieving consistent sub-**100ms** latency with **99.9%** uptime.
 - Developed **automated DevOps workflows** for dynamic **Sharder/Miner** management and view-change protocols, boosting developer productivity by **90%** and reducing manual intervention.
 - Enhanced **multi-wallet concurrency** handling with unified **GoSDK**, supporting **10K+ concurrent** transactions and improving load test performance by **300%**.
- **Oppia Foundation (incl. GSoC) [Merged PRs]** Mountain View, CA (Remote)
OSS Contributor & GSoC - Kotlin, GitHub Actions, JUnit, Gradle, Bazel, Android SDK Jun 2021 - Sep 2022
 - Contributed to **Gradle to Bazel** migration, achieving **10x** improvement in build speeds.
 - Designed robust **CI/CD infrastructure** (GSoC) with automated build triggers and deployment to **Google Drive, GCP Bucket**, and **Google Play Store** tracks via custom Kotlin scripts and GitHub Actions.
 - Developed modular **MVVM/MVP** architectures with **tablet support** for multiple screen sizes.
 - Authored **70+ Unit Tests** and **20+ E2E tests** using Espresso, Robolectric, and UIAutomator.
- **Gnome Foundation** Remote
Open Source Contributor - Tech Stack: C, XML, GTK+ Sep 2021 - Feb 2022
 - Enhanced **Nautilus file manager** with new path-bar features, improving user efficiency by **20%**.
 - Implemented **drag-and-drop support** and keyboard shortcuts for seamless file organization.

PROJECTS

- **Distributed Key-Value Store (Raft-based):** Implemented a fault-tolerant KV store with leader election, log replication, and state machine application using the Raft consensus algorithm.

ACHIEVEMENTS

- Practised over **700+** DSA problems on Leetcode.
- Member of **Oppia Foundation** for **2 years**.
- Completed **Cloud Engineering tracks** 30 Days of Google Cloud Program.

TECHNICAL SKILLS

- **Programming & Scripting:** Java, Go, Kotlin, Python, JavaScript, SQL, Bash, C/C++
- **Backend Frameworks:** Spring Boot, Ruby on Rails, Gin, Express.js, REST APIs, GraphQL, Microservices
- **DevOps & Cloud:** AWS (RDS), Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Helm, Prometheus/Grafana/Loki
- **Databases & Testing:** MySQL, PostgreSQL, MongoDB, Redis, ClickHouse, Query Optimization, JUnit, Mockito
- **Tools & Libraries:** Git, Apache Kafka, Debezium, CDC, gh-ost, Semgrep, Gradle, Bazel, System Design